

TWO-STAGE GROWTH MODEL

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NET PP&E, DEPRECIATION, CAPITAL EXPENDITURE

Three lines

$$\text{Net PP\&E}_t = k \times \text{Sales}_{t+1}$$

$$\text{Depreciation}_t = d \times \text{Net PP\&E}_{t-1}$$

$$\text{CapEx}_t = \text{Net PP\&E}_t - \text{Net PP\&E}_{t-1} + \text{Depreciation}_t$$

Net PP&E and the depreciation rate are assumptions. Capital expenditure is computed from them.

Capital expenditure as the residual

The asset base is the assumption

Capacity is in place before the sales it serves, so the balance at the end of year t carries year $t + 1$'s sales.

Depreciation follows the balance

A percentage of the opening net balance, not a percentage of sales.

Capital expenditure is the residual

It funds the increase in the balance and replaces what was depreciated during the year.

The zero-growth check

If the balance never changes, capital expenditure equals depreciation.

A FIVE-YEAR PROJECT

Assumptions

ITEM	ASSUMPTION
Sales, year 1	32.00
Sales growth	8% a year
EBITDA margin	25%
Net PP&E	50% of next year's sales
Depreciation	10% of opening net PP&E
Net working capital	10% of next year's sales
Tax rate	21%
Discount rate	12%
Life	5 years, assets sold at book value

The capital schedule

\$ MILLIONS	0	1	2	3	4	5
Sales	—	32.00	34.56	37.32	40.31	43.54
Net PP&E	16.00	17.28	18.66	20.16	21.77	19.59
Depreciation	—	1.60	1.73	1.87	2.02	2.18
Capital expenditure	16.00	2.88	3.11	3.36	3.63	—
Net working capital	3.20	3.46	3.73	4.03	4.35	—

Year 5 buys nothing: there is no year 6 to build for. The closing 19.59 is the year-4 balance less year-5 depreciation. Figures are rounded to 0.01.

The income statement

\$ MILLIONS	1	2	3	4	5
Sales	32.00	34.56	37.32	40.31	43.54
EBITDA	8.00	8.64	9.33	10.08	10.88
Depreciation	1.60	1.73	1.87	2.02	2.18
EBIT	6.40	6.91	7.46	8.06	8.71
Taxes at 21%	1.34	1.45	1.57	1.69	1.83
Net operating profit after tax	5.06	5.46	5.90	6.37	6.88

Add depreciation back

\$ MILLIONS	1	2	3	4	5
Net operating profit after tax	5.06	5.46	5.90	6.37	6.88
Depreciation	1.60	1.73	1.87	2.02	2.18
Operating cash flow	6.66	7.19	7.76	8.38	9.06

Depreciation lowered the tax bill without taking any cash, so it goes back in.

Free cash flow

\$ MILLIONS	0	1	2	3	4	5
Operating cash flow	—	6.66	7.19	7.76	8.38	9.06
Capital expenditure	(16.00)	(2.88)	(3.11)	(3.36)	(3.63)	—
Change in working capital	(3.20)	(0.26)	(0.28)	(0.30)	(0.32)	4.35
Sale at book value, untaxed	—	—	—	—	—	19.59
Free cash flow	(19.20)	3.52	3.80	4.11	4.43	33.00

NPV and IRR

\$11.44m

NPV AT 12%

26.3%

IRR

\$19.20m

INVESTED AT DATE 0

With constant ratios and a sale at book, the IRR is 26.33%, and so is year-1 operating profit after tax over the 19.20 of capital.

THE PROJECT THAT DOES NOT END

Drop the sale, keep the ratios

No sale

The 19.59 of book value stays in the business.

Year 5 keeps investing

Capital expenditure of 3.92 and another 0.35 of working capital, building for year 6.

\$ MILLIONS	1	2	3	4	5
Free cash flow	3.52	3.80	4.11	4.43	4.79
Share of sales	11.0%	11.0%	11.0%	11.0%	11.0%

Every line is a multiple of sales

Constant growth and constant ratios mean free cash flow grows at the sales rate forever.

$$V_0 = \frac{FCF_1}{r - g} = \frac{3.52}{0.12 - 0.08} = 88.00$$

Against 19.20 of capital at date 0, an NPV of 68.80 — against 11.44 when the same business was sold at book in year 5.

THE FIRM

Nothing is bought at date 0

The 16.00 of net PP&E and 3.20 of working capital are already there. No outflow at date 0, so the 88.00 is the value of the firm.

\$88.00m

VALUE OF THE FIRM

11.0x

EBITDA

4.6x

INVESTED CAPITAL

TWO-STAGE GROWTH

The first three years, and then 8%

The same company, with sales that reach 8% only in year 4: 25% growth in year 2, 15% in year 3, and a margin that climbs to 25% over the same three years. It holds 6.00 of cash and 18.00 of debt, against 4.0 million shares.

Years 1 to 3

Sales grow 25% and then 15%. The EBITDA margin climbs from 18% to 21% to 23%.

Year 4 onward

Sales grow 8% a year and the margin holds at 25%. The three ratios are fixed throughout.

Assumptions

ITEM	ASSUMPTION
Sales, year 1	32.00
Sales growth	25%, 15%, then 8% from year 4
EBITDA margin	18%, 21%, 23%, then 25% from year 4
Net PP&E	50% of next year's sales
Depreciation	10% of opening net PP&E
Net working capital	10% of next year's sales
Tax rate	21%
WACC	12%
Cash, debt, shares	6.00, 18.00, 4.0 million

The capital schedule

\$ MILLIONS	0	1	2	3	4	5
Sales	—	32.00	40.00	46.00	49.68	53.65
Next year's sales	32.00	40.00	46.00	49.68	53.65	57.95
Net PP&E	16.00	20.00	23.00	24.84	26.83	28.97
Depreciation	—	1.60	2.00	2.30	2.48	2.68
Capital expenditure	—	5.60	5.00	4.14	4.47	4.83
Net working capital	3.20	4.00	4.60	4.97	5.37	5.79

The date-0 balances are what the company already owns. Figures are rounded to 0.01.

Three of these lines are plotted, with sliders, at mgmt648.kerryback.com/ppe-explorer.html.

The income statement

\$ MILLIONS	1	2	3	4	5
Sales	32.00	40.00	46.00	49.68	53.65
EBITDA	5.76	8.40	10.58	12.42	13.41
Depreciation	1.60	2.00	2.30	2.48	2.68
EBIT	4.16	6.40	8.28	9.94	10.73
Taxes at 21%	0.87	1.34	1.74	2.09	2.25
Net operating profit after tax	3.29	5.06	6.54	7.85	8.48

Add depreciation back

\$ MILLIONS	1	2	3	4	5
Net operating profit after tax	3.29	5.06	6.54	7.85	8.48
Depreciation	1.60	2.00	2.30	2.48	2.68
Operating cash flow	4.89	7.06	8.84	10.33	11.16

Depreciation lowered the tax bill without taking any cash, so it goes back in.

Free cash flow

\$ MILLIONS	1	2	3	4	5
Operating cash flow	4.89	7.06	8.84	10.33	11.16
Capital expenditure	(5.60)	(5.00)	(4.14)	(4.47)	(4.83)
Increase in working capital	(0.80)	(0.60)	(0.37)	(0.40)	(0.43)
Free cash flow	(1.51)	1.46	4.33	5.46	5.90
Change from prior year	—	n/m	197.6%	26.1%	8.0%
Share of sales	(4.7%)	3.6%	9.4%	11.0%	11.0%

Changes are computed before rounding.

TERMINAL VALUE

Free cash flow in year 5

Free cash flow reaches 11.0% of sales in year 4 and stays there: the margin and all three ratios are fixed from year 4 on. Sales grow 8% in year 5 and every year after, so free cash flow does too.

Year 4 is the first year at the steady ratio. Year 5 is the first year that grows at 8%, which is why the terminal value sits at the end of year 4.

The terminal value

$$TV_4 = \frac{FCF_5}{r - g} = \frac{5.90}{0.12 - 0.08} = 147.55$$

Twenty-five times year-5 free cash flow, at a four-point spread between r and g .

Discount it all back

\$ MILLIONS	1	2	3	4	TV AT 4, YEAR 5 ON
Cash flow	(1.51)	1.46	4.33	5.46	147.55
Discount factor at 12%	0.893	0.797	0.712	0.636	0.636
Present value	(1.35)	1.16	3.08	3.47	93.77

\$100.14m
ENTERPRISE VALUE

94%
IN THE TERMINAL VALUE

Where the 8% comes from

$$g = \text{ROIC} \times \text{reinvestment rate}$$

ROIC

$$8.48/32.20 = 26.3\%$$

Reinvestment rate

$$(4.83 + 0.43 - 2.68)/8.48 = 30.4\%$$

Product

$$26.3\% \times 30.4\% = 8.0\%$$

Year-5 figures, against net PP&E plus working capital of 32.20 at the end of year 4.

CASH AND DEBT

From enterprise value to equity

\$ MILLIONS

Enterprise value	100.14
Plus cash	6.00
Less debt	(18.00)
Equity value	88.14
Shares outstanding	4.0 million
Value per share	\$22.03

Two points about the equity value calculation

Cash is not working capital

The 10% ratio covers receivables, inventory and payables. Cash is added separately, at face value.

Interest is not in free cash flow

The cash flows are unlevered. Debt enters through the WACC and again in the equity value calculation, never in the forecast.

EXERCISE

Value this company

The forecast

Sales of 20.00 in year 1, growing 30%, then 18%, then 6% from year 4 on. EBITDA margin 15%, 18%, 20%, then 22% from year 4 on.

The ratios

Net PP&E 65% of next year's sales.

Depreciation 12% of opening net PP&E.

Working capital 12% of next year's sales.

Tax 21%, WACC 12%.

- 1 Free cash flow for years 1 to 4
- 2 Terminal value at the end of year 4, and the enterprise value
- 3 Equity value per share, with cash of 4.00, debt of 15.00, and 2.5 million shares